

A sharp collapsed-fibre likelihood classification for a four-visible restricted Boltzmann machine

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[Code and formal proofs](#)

Abstract

For pure four-parity data and a four-visible, one-hidden spin RBM, collapsed parameters are local likelihood maxima precisely below an explicit $\tanh^2$ threshold. At equality and beyond they are saddles in the full parameter space; every collapsed point remains globally suboptimal and non-strict.

The model and full-space theorem

For $s \in \{-1, 1\}^4$ and $0 < |\rho| \leq 1$, take data distribution $Q_\rho(s) = (1 + \rho s_1 s_2 s_3 s_4) / 16$. The one-hidden spin RBM has normalized probabilities proportional to $\exp(b \cdot s) \cosh(c + w \cdot s)$, with $b, w \in \mathbb{R}^4$ and $c \in \mathbb{R}$. Its expected log likelihood is ℓ_ρ . The collapsed fibre consists of $\theta_0 = (0, c_0, 0)$, which all represent the uniform visible distribution.

Every affine directional derivative at θ_0 is zero. The sharp theorem says θ_0 is a local maximum in the full nine-dimensional parameter space if and only if $\tanh^2(c_0) < (3 + 2|\rho|) / (3 + 6|\rho|)$. At equality or above, every full parameter neighbourhood contains both a strictly higher and a strictly lower likelihood point. Every θ_0 also has some finite RBM with strictly better likelihood and distinct equal-likelihood points arbitrarily nearby.

Why higher-order analysis is needed

The collapsed parametrization is singular. Ordinary stationarity and a quadratic Hessian test miss the parity interaction, whose first nonzero signal appears at higher order. The proof expands $\log \cosh$ and decomposes the visible functions into Walsh/parity components. It controls nuisance visible biases and normalizing terms by quantitative Jensen and likelihood-domination estimates.

The quartic geometry produces the threshold. An independent ordinary fourth-order derivation agrees with the displayed expression, but that calculation alone does not prove a full neighbourhood maximum. The source's `StableRegion` and `LocalMaximum` lemmas supply uniform control over arbitrary small parameter perturbations, rather than just along one symmetric ray.

Outside the threshold, a quartic escape direction gives higher likelihood and another direction gives lower likelihood. At equality, moving along the likelihood-flat collapsed fibre reaches nearby unstable parameters, placing both behaviours in every neighbourhood of the boundary point. A strictly signed quartic test alone would not establish this equality case.

Non-strictness and global suboptimality

Changing the hidden bias while keeping $b=w=0$ leaves the visible distribution uniform, which creates a flat direction and makes collapsed local maxima non-strict. A suitable finite coupled RBM captures part of the nonzero parity signal and improves the likelihood. Thus a collapsed local maximum can be nonglobal even though it has nearby points of equal value.

The theorem concerns parameter-space likelihood only. It does not assert a distribution-space maximum, an attracting training basin, strict trapping, EM convergence or a lower bound on optimization time. It fixes exactly four visible spins, one hidden unit and pure parity data. Ordinary explorations for larger dimensions or general four-wise-uniform data are outside the selected formal endpoint.

Prior setting and formal scope

Collapsed stationarity, singular-mixture critical fibres and RBM interaction formulas are prior mathematics. In the Fukumizu-Akaho-Amari comparison the second-derivative splitting matrix vanishes. The refinement here is the exact higher-order threshold with its full-neighbourhood and equality-boundary statements.

Formal verification

Lean 4.33.1 checked the 15-source closure and all three selected declarations `sharp_parity_classification`, `collapsed_local_max_iff` and `nonstable_parameter_saddle`, using only standard kernel axioms. The result covers exactly four visible spins, one hidden unit and pure four-parity data, including the equality boundary. Broader RBMs, optimization-time claims and training convergence are outside this verification. Pinned sources and audit records are supplied in the companion repository.

AI assistance. Codex assisted literature checking, editorial preparation and analysis of the supplied formal artifacts. The human authors are responsible for checking the final mathematical claims, references and attribution.

ParityFour.sharp_parity_classification

```
theorem sharp_parity_classification (p c₀ : ℝ) (hp : 0 < |p|) (hp1 : |p| ≤ 1) :
(∀ (b : Fin 4 → ℝ) (d : ℝ) (w : Fin 4 → ℝ),
HasDerivAt (fun t => objective p (affineParameterRay c₀ b d w t)) 0 0) ∧
(IsLocalMax (objective p) (nullParameter c₀) ↔
Real.tanh c₀ ^ 2 < (3+2*|p|)/(3+6*|p|)) ∧
((3+2*|p|)/(3+6*|p|) ≤ Real.tanh c₀ ^ 2 → ParameterSaddle p (nullParameter c₀)) ∧
(∃ θ : Parameters, objective p (nullParameter c₀) < objective p θ) ∧
(∀ U : Set Parameters, U ∈ nhds (nullParameter c₀) →
∃ θ : Parameters, θ ∈ U ∧ θ ≠ nullParameter c₀ ∧
objective p θ = objective p (nullParameter c₀))
```

Source: proofs/opdp89-parity-sharp/SharpClassification.lean:40.

Source SHA-256: 15ef1ad86c585c87e7ac4fabdc3a58702c915e69dbf6aa80d701258bc9efc602

ParityFour.collapsed_local_max_iff

```
theorem collapsed_local_max_iff (p c₀ : ℝ) (hp : 0 < |p|) (hp1 : |p| ≤ 1) :  
IsLocalMax (objective p) (nullParameter c₀) ↔  
Real.tanh c₀ ^ 2 < (3+2*|p|)/(3+6*|p|)
```

Source: proofs/opdp89-parity-sharp/SharpClassification.lean:22.

Source SHA-256: 15ef1ad86c585c87e7ac4fabdc3a58702c915e69dbf6aa80d701258bc9efc602

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